

Jason M. Merlo

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Research Interests

Wireless electrical state coordination (time, frequency, phase); software-defined radios (SDRs); distributed radar and communications systems; interferometric radar arrays; synthetic aperture radar (SAR); automotive radar.

Education

Ph.D. Electrical Engineering

August 2025

Michigan State University

East Lansing, MI

Dissertation: *Wirelessly Coordinated Open-Loop Coherent Distributed Antenna Arrays Using Software-Defined Radio*

Committee: Jeffrey Nanzer (Chair/Advisor), Premjeet Chahal, Hayder Radha, Huacheng Zeng

B.S. Computer Engineering

December 2018

Michigan State University

East Lansing, MI

Research Experience

Assistant Professor (Research), Fixed-Term

Oct. 2025—Present

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

Distributed Electromagnetics Theory and Application (DELTA) Group

Oct. 2025—Present

- Conducted research on wireless radio coordination and distributed radar and communication techniques using software-defined radios (SDRs)
- Lead software planning and maintenance for internal group software libraries and frameworks in Python and C++ for SDRs
- Mentored graduate students in digital signal processing for sensing wireless coordination on SDRs
- Contributed technical content to funding proposals

Laboratory for Electromagnetics and mm-Wave Antennas (LEmmA)

Oct. 2025—Present

- Developed multi-platform near-field antenna measurement processing software using Python and Qt 6 GUI framework for interactive plotting and data manipulation
- Implemented high performance planar-near field to far field transformation algorithms using non-uniform fast Fourier transform for efficient data processing

Professional Aide (Research)

Aug. 2025—Oct. 2025

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

Distributed Electromagnetics Theory and Application (DELTA) Group

- Continued research on wirelessly coordinated coherent distributed antenna arrays and software-defined radio

Graduate Research Assistant

Jan. 2019—Aug. 2025

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

Distributed Electromagnetics Theory and Application (DELTA) Group

- Research topics included:
 - Wirelessly coordinated SDR-based coherent distributed antenna arrays

- Automotive communication waveform reuse for SAR imaging for scene mapping and localization
- Distributed aperture interferometric radar for direct multi-dimensional target tracking
- Developed and maintained internal Python (Numba) and C++ software packages for software-defined radio, radar, and distributed array coordination, simulation, and real-time signal processing
- Developed and maintained the “DELTA Distributed Array Framework,” a distributed computing framework for managing deployment and synchronization of libraries and applications in ephemeral, ad-hoc distributed, networked compute arrays for rapid distributed array prototyping and development
- Designed and fabricated a low cost integrated 5.8 GHz FMCW radar PCB for drone detection and tracking
 - Implemented a high performance real-time data processing and monitoring dashboard in Python and PyQt
- Managed software-defined radios and computing resources for the research group
- Contributed technical content to funding proposals
- Mentored and trained new group members on lab equipment use and care, and measurement best practices

Summer Research scholar

May 2023—Aug. 2023

Lawrence Livermore National Laboratory

Livermore, CA

- Continuation of doctoral research related to the high accuracy time, frequency, and phase coordination of wirelessly coordinated coherent distributed antenna arrays.

Undergraduate Research Assistant

May 2018—Dec. 2018

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

Distributed Electromagnetics Theory and Application (DELTA) Group

- Designed, simulated, and fabricated miniaturized Doppler radar baseband amplifier printed circuit board
- Designed and implemented real-time radar data acquisition, processing, and visualization package for Python using Qt 5 and PyQtGraph
- Investigated various approaches to reduce tracking error and drift using Doppler-only radar array measurements
- Assisted with measurements and fabrication of microwave tomography device for forestry applications

Teaching and Mentoring Experience

Graduate Student Mentor

Sept. 2020—Oct. 2025

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

Distributed Electromagnetics Theory and Application (DELTA) Group

- Mentored undergraduate and graduate research assistants on research related to software defined radios, wireless distributed coordination, antenna arrays, and microwave system fabrication and testing
- Mentored new students in software architecture and development best practices using Python, C++, and GNU Radio; git version control; reproducible software environments using Docker and Conda/Pixi; software testing using PyTest; and GitLab CI/CD pipelines

Graduate Student Mentor

Jan. 2019—June 2021

Michigan State University, Electrical and Computer Engineering

East Lansing, MI

General Motors / SAE AutoDrive Challenge Competition

- Mentored incoming students in:
 - Software development using C++ and Python using ROS and software architecture best practices
 - Automotive electrical harness assembly and automotive electronics best practices
 - Performing design failure mode effects analysis (DFMEA) and mitigations
- Mentored three senior design teams in:
 - Radar placement and enclosure design (sensor placement analysis, DFMEA)
 - Power distribution and safety system (system design, DFMEA, and PCB layout best practices)
 - High-reliability sensor interfacing system (system design, DFMEA, and PCB layout best practices)

Undergraduate Teaching Assistant

*Michigan State University, Computer Science and Engineering
Introduction to Programming I in Python (CSE 231)*

Jan. 2017—May 2017

East Lansing, MI

- Led weekly lab session; compiled and presented a weekly skills recap presentation
- Worked weekly help room and answered questions via online forum
- Graded lab section's weekly project assignments

Awards and Fellowships

2nd Place, Student Paper Contest, IEEE International Symposium on Phased Array Systems & Technology
IEEE Boston Section **Oct. 2024**

- For the paper "Fully Wireless Collaborative Beamforming Using A Three-Element Coherent Distributed Phased Array"

Finalist, Student Paper Contest, IEEE International Symposium on Antennas and Propagation **July 2024**
IEEE Antennas and Propagation Society

- For the paper "Distributed Interferometric Radar for Radial and Angular Velocity Measurement"

Finalist, Student Paper Contest, IEEE/MTT-S International Microwave Symposium (IMS) **June 2024**
IEEE Microwave Theory and Technology Society

- For the paper "Fully Wireless Coherent Distributed Phased Array System for Networked Radar Applications"

Selected Papers From IMS 2024, IEEE Microwave and Wireless Technology Letters **June 2024**
IEEE Microwave Theory and Technology Society

- For the paper "Fully Wireless Coherent Distributed Phased Array System for Networked Radar Applications"

IEEE MTT-S Graduate Fellowship Recipient **2023—2024**
IEEE Microwave Theory and Technology Society

- For the support of research towards "High Accuracy Wireless Distributed Coherent Array Synchronization"

Young Scientist Awardee, URSI General Assembly **Aug. 2023**
International Union of Radio Science (URSI)

- For the paper "Picosecond Non-Line-of-Sight Wireless Time and Frequency Synchronization for Coherent Distributed Aperture Antenna Arrays"

Honorable Mention, IEEE International Symposium on Antennas and Propagation Student Paper Contest
IEEE Antennas and Propagation Society **July 2023**

- For the paper "Wireless Time and Phase Alignment for Wideband Beamforming in Distributed Phased Arrays"

First Place, IEEE/MTT-S International Microwave Symposium (IMS) Student Paper Contest **June 2023**
IEEE Microwave Theory and Technology Society

- For the paper "High Accuracy Wireless Time-Frequency Transfer For Distributed Phased Array Beamforming"

Departmental Finalist (Electrical Eng.), Fitch H. Beach Award for Outstanding PhD Research in Engineering **Mar. 2023**

Michigan State University, College of Engineering

Second Place, IEEE AP-S Student Design Competition **July 2020**
IEEE Antennas and Propagation Society

- For the project "A Low-Cost 5.8 GHz FMCW Radar For Drone Tracking"

Third Place, Electrical Engineering Capstone Design Day **April 2018**
Michigan State University, Electrical and Computer Engineering

- For the projects "Lane Following and Obstacle Avoidance" and "Autonomous Vehicle Safety & Integrity Monitor (SIM)" on the CANVAS/SOAR Chevrolet Bolt AV

- For the project "Doppler Radar Pedestrian Classifier"

Publications

Real-Time High-Accuracy Digital Wireless Time, Frequency, and Phase Calibration For Coherent Distributed Antenna Arrays **Dec. 2025**

Jason M. Merlo, Samuel Wagner, John Lancaster, and Jeffrey A. Nanzer

IEEE Transactions on Microwave Theory and Techniques

doi: 10.1109/TMTT.2025.3639461

Distributed Beamforming Using Decentralized Time Synchronization in a Six-Element Array **Sept. 2025**

Naim Shandi, **Jason M. Merlo**, and Jeffrey A. Nanzer

IEEE Journal of Microwaves

doi: 10.1109/JMW.2025.3601036

Three-Dimensional Millimeter-Wave Imaging Using Active Incoherent Fourier Processing and Pulse Compression (Preprint) **Jul. 2025**

Jorge R. Colon-Berrios, **Jason M. Merlo**, and Jeffrey A. Nanzer

doi: 10.48550/arXiv.2507.07239

Collaborative Beamforming for Communication Applications Using a Two-Element Fully-Wireless Open-Loop Coherent Distributed Array (Preprint) **Jun. 2025**

Jason M. Merlo, and Jeffrey A. Nanzer

doi: 10.48550/arXiv.2506.13014

Decentralized Picosecond Synchronization for Distributed Wireless Systems **Oct. 2024**

Naim Shandi, **Jason M. Merlo**, and Jeffrey A. Nanzer

IEEE Transactions on Communications

doi: 10.1109/TCOMM.2024.3480993

Fully Wireless Coherent Distributed Phased Array System for Networked Radar Applications **Mar. 2024**

Jason M. Merlo, Samuel Wagner, John Lancaster, and Jeffrey A. Nanzer

IEEE Microwave and Wireless Technology Letters

doi: 10.1109/LMWT.2024.3375085

Array Phase Center Dynamics Using Spatial Amplitude Modulation for High Efficiency Secure Wireless Communication **Aug. 2023**

Jacob R. Randall, **Jason M. Merlo**, Amer Abu Arisheh, and Jeffrey A. Nanzer

IEEE Antennas and Wireless Propagation Letters

doi: 10.1109/TAP.2023.3323141

A Dynamic Array Using Spatial Amplitude Modulation with an Asymmetric Wilkinson Power Divider for Secure Wireless Applications **Aug. 2023**

Jacob R. Randall, Amer Abu Arisheh, **Jason M. Merlo**, and Jeffrey A. Nanzer

IEEE Antennas and Wireless Propagation Letters

doi: 10.1109/LAWP.2023.3310911

Coherent Distributed Bistatic Radar Using Wireless Frequency Syntonization and Internode Ranging

Anton Schlegel, **Jason M. Merlo**, and Jeffrey A. Nanzer

July 2023

IEEE Microwave and Wireless Technology Letters

doi: 10.1109/LMWT.2023.3283918

Design of a Single-Element Dynamic Antenna for Secure Wireless Applications **June 2023**

Amer Abu Arisheh, **Jason M. Merlo**, and Jeffrey A. Nanzer

IEEE Microwave and Wireless Technology Letters

doi: 10.1109/TAP.2023.3288013

Multiobjective Distributed Array Beamforming in the Near Field Using Wireless Syntonization **Feb. 2023**

Ahona Bhattacharyya, **Jason M. Merlo**, Serge R. Mghabghab, Anton Schlegel, and Jeffrey A. Nanzer

IEEE Transactions on Antennas and Propagation

doi: 10.1109/LMWT.2022.3231183

Wireless Picosecond Time Synchronization for Distributed Antenna Arrays **Dec. 2022**

Jason M. Merlo, Serge R. Mghabghab, and Jeffrey A. Nanzer

IEEE Transactions on Microwave Theory and Techniques

doi: 10.1109/TMTT.2022.3227878A

- Microwave Tomography System Using Time-Reversal Imaging for Forestry Applications** **Oct. 2022**
 Saptarshi Mukerjee, John Doroshewitz, **Jason M. Merlo**, Christopher Oakley, Lalita Udpa, David MacFarlane, Emily Huff, and Jeffrey A. Nanzer
IEEE Journal of Microwaves doi: 10.1109/JMW.2022.3199194
- A C-Band Fully Polarimetric Automotive Synthetic Aperture Radar** **Dec. 2021**
Jason M. Merlo and Jeffrey A. Nanzer
IEEE Transactions on Vehicular Technology doi: 10.1109/TVT.2021.3138348
- A Point Target Model for Interferometric Radar Angular Velocity Estimation** **Dec. 2021**
 Eric Klinefelter, **Jason M. Merlo**, Hayder Radha, and Jeffrey A. Nanzer
IEEE Transactions on Microwave Theory and Techniques doi: 10.1109/TMTT.2021.3136265
- Distributed Antenna Array Dynamics for Secure Wireless Communication** **Dec. 2021**
 Sean M. Ellison, **Jason M. Merlo**, and Jeffrey A. Nanzer
IEEE Transactions on Antennas and Propagation doi: 10.1109/TAP.2021.3137449
- Three Dimensional Velocity Measurement Using a Dual Axis Millimeter-Wave Interferometric Radar** **Nov. 2021**
Jason M. Merlo, Eric Klinefelter and Jeffrey A. Nanzer
IEEE Transactions on Microwave Theory and Techniques doi: 10.1109/TMTT.2021.3124251
- A Multiple Baseline Interferometric Radar for Multiple Target Angular Velocity Measurement** **Aug. 2021**
Jason M. Merlo, Eric Klinefelter, Stavros Vakalis, and Jeffrey A. Nanzer
IEEE Microwave and Wireless Components Letters doi: 10.1109/LMWC.2021.3079842

Conference Papers and Presentations

* Denotes Presenter

- All-Digital Wirelessly Coordinated Phased Array Collaborative Beamforming Using High Accuracy Time Synchronization (Presentation Only)** **July 2025**
Jason M. Merlo*, and Jeffrey A. Nanzer Ottawa, Canada
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI)
- Impact of Localization and Synchronization Errors on One-Dimensional Distributed Interferometric Microwave Imaging** **July 2025**
 Derek Luzano*, **Jason M. Merlo**, and Jeffrey A. Nanzer Ottawa, Canada
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI) doi: 10.1109/AP-S/CNC-USNC-URSI55537.2025.11266039
- Wireless Time Synchronization for Distributed Phased Arrays Using Frequency Hopping** **July 2025**
 Naim Shandi*, **Jason M. Merlo**, and Jeffrey A. Nanzer Ottawa, Canada
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI) doi: 10.1109/AP-S/CNC-USNC-URSI55537.2025.11266731
- Three-Dimensional Fourier Domain Millimeter-Wave Imaging Using Incoherent Active Illumination and Pulse Compression** **June 2025**
 Jorge R. Colon-Berrios*, **Jason M. Merlo**, and Jeffrey A. Nanzer San Francisco, CA
2025 IEEE/MTT-S International Microwave Symposium doi: 10.1109/IMS40360.2025.11103855
- Fully Wireless Collaborative Beamforming Using A Three-Element Coherent Distributed Phased Array**
2nd Place, Student Paper Contest **Sept. 2024**
Jason M. Merlo*, Naim Shandi, Matthew Dula, Ahona Bhattacharyya, and Jeffrey A. Nanzer
 IEEE International Symposium on Phased Array Systems & Technology (PAST) Boston, Massachusetts, USA

- Distributed Interferometric Radar for Radial and Angular Velocity Measurement** **July 2024**
Finalist, Student Paper Contest Florence, Italy
Jason M. Merlo*, and Jeffrey A. Nanzer
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI) doi: 10.1109/AP-S/INC-USNC-URSI52054.2024.10686149
- High Accuracy Decentralized Time Synchronization Using SNR Based Weighting** **July 2024**
 Naim Shandi*, **Jason M. Merlo**, and Jeffrey A. Nanzer Florence, Italy
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI) doi: 10.1109/AP-S/INC-USNC-URSI52054.2024.10686155
- A Distributed Microwave Correlation Interferometer for Fourier Domain Imaging Using Wireless Time and Frequency Coordination** **July 2024**
 Derek Luzano*, **Jason M. Merlo**, Daniel Chen, Jorge R. Colon-Berrios, Ahona Bhattacharyya, and Jeffrey A. Nanzer Florence, Italy
IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI) doi: 10.1109/AP-S/INC-USNC-URSI52054.2024.10686889
- Fully Wireless Coherent Distributed Phased Array System for Networked Radar Applications** **June 2024**
Finalist, Student Paper Contest Washington, D.C., USA
Jason M. Merlo*, Samuel Wagner, John Lancaster, and Jeffrey A. Nanzer
IEEE Microwave and Wireless Technology Letters / MTT-S International Microwave Symposium (IMS) doi: 10.1109/LMWT.2024.3375085
- High Accuracy Wireless Timing Synchronization Using Software Defined Radios** **Sept. 2023**
 Ahona Bhattacharyya*, **Jason M. Merlo**, and Jeffrey A. Nanzer Berlin, Germany
20th European Radar Conference (EuRAD), pp. 331–334 doi: 10.23919/EuRAD58043.2023.10289274
- High Accuracy Wireless Timing Synchronization Using Software Defined Radios** **Aug. 2023**
Jason M. Merlo*, and Jeffrey A. Nanzer Tempe, Arizona, USA
GNU Radio Conference (GRCon)
- Picosecond Non-Line-of-Sight Wireless Time and Frequency Synchronization for Coherent Distributed Aperture Antenna Arrays** **Aug. 2023**
Young Scientist Awardee Sapporo, Japan
Jason M. Merlo*, and Jeffrey A. Nanzer
XXXVth General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS) doi: 10.23919/URSIGASS57860.2023.10265506
- Wireless Time and Phase Alignment for Wideband Beamforming in Distributed Phased Arrays** **July 2023**
Honorable Mention, Student Paper Contest Portland, Oregon, USA
Jason M. Merlo*, and Jeffrey A. Nanzer
IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, pp. 365–366 doi: 10.1109/USNC-URSI52151.2023.10238024
- Sub-Millimeter Ranging Accuracy for Distributed Antenna Arrays Using Two-Way Time Transfer** **July 2023**
 Naim Shandi*, **Jason M. Merlo**, and Jeffrey A. Nanzer Portland, Oregon, USA
IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, pp. 517–518 doi: 10.1109/IMS37964.2023.10188022
- High Accuracy Wireless Time-Frequency Transfer For Distributed Phased Array Beamforming** **June 2023**
First Place, Student Paper Contest San Diego, California, USA
Jason M. Merlo*, Anton Schlegel, and Jeffrey A. Nanzer
IEEE/MTT-S International Microwave Symposium (IMS), pp. 109–112 doi: 10.1109/IMS37964.2023.10188022

Multi-Pass Automotive Synthetic Aperture Radar Image Fusion (Invited)**May 2023****Jason M. Merlo*** and Jeffrey A. Nanzer

San Antonio, Texas, USA

IEEE Radar Conference

doi: 10.1109/RadarConf2351548.2023.10149757

High Accuracy Wireless Time Synchronization for Distributed Antenna Arrays**July 2022****Jason M. Merlo*** and Jeffrey A. Nanzer

Denver, Colorado, USA

IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, pp.

1966–1967

doi: 10.1109/AP-S/USNC-URSI47032.2022.9887217

A Dynamic Pattern Dipole Antenna for Secure Wireless Communications**July 2022**Amer Abu Arisheh*, **Jason M. Merlo**, and Jeffrey A. Nanzer

Denver, Colorado, USA

IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, pp.

1470–1471

doi: 10.1109/AP-S/USNC-URSI47032.2022.9887173

Joint Measurement of Target Angle and Angular Velocity Using Interferometric Radar with FM Waveforms**Sept. 2020****Jason M. Merlo***, Eric Klinefelter, and Jeffrey A. Nanzer

Florence, Italy (Virtual)

IEEE Radar Conference

doi: 10.1109/RadarConf2043947.2020.9266419

A Dual-Axis Interferometric Radar for Instantaneous 2D Angular Velocity Measurement**July 2020****Jason M. Merlo***, Eric Klinefelter, and Jeffrey A. Nanzer

Montréal, Canada (Virtual)

IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, pp.

1661–1662

doi: 10.1109/IEEECONF35879.2020.9330294

A Microwave Tomography System Using Time-Reversal Imaging**July 2019**John Doroshewitz*, **Jason M. Merlo**, Christopher Oakley, Lalita Udpa, Jeffrey A. Nanzer, David MacFarlane,

Emily Huff, and Saptarshi Mukherjee

Atlanta, Georgia, USA

2019 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting

doi: 10.1109/APUSNCURSINRSM.2019.8889157

Collegiate Design Competitions**General Motors / SAE AutoDrive Challenge Competition****Aug. 2017—June 2021***Project Manager / Team Co-Lead, Electrical Sub-Team Leader*

- Leading overall vehicle development and electrical hardware development during four-year competition

Graduate Student Advisor: Perception Sub-Team

- Guiding efforts to develop, implement, and test object detection and lane-finding algorithms using camera, lidar, and radar using ROS in C++ and Python

IEEE Antennas and Propagation Society Student Design Competition**Nov. 2019—July 2020***Drone Detection and Tracking Radar*

- Led design, simulation, and assembly of PCB for a low-cost, 4-channel, 5.8 GHz FMCW drone detection and tracking radar system
- Collaborated with others to create a real-time tracking and data visualization dashboard using Python and PyQt/PyQtGraph
- Placed second overall and presented at the 2020 IEEE International Symposium on Antennas and Propagation and North American Radio Science Meeting
- Open sourced hardware, software source code and instructions – available at: <https://gitlab.msu.edu/delta/aps2020-competition>

NHTSA Safety Technology Design Competition

Sept. 2016—July 2017

Doppler Radar Pedestrian Classifier

- Led team of students to designed, assemble, and test a portable X-band CW radar for pedestrian detection
- Placed in the top six semi-finalist teams and presented at the “Enhanced Safety of Vehicles” conference in Detroit, July 2017

MSU Solar Racing Team

Sept. 2014—Mar. 2015

Electrical Sub-Team Member

- Wrote C library for SPI interface CAN controller and data-logging software for wireless communications via XBee radio.
- Designed PCB for MSP430 development board used to test CAN system, 7-segment dashboard displays driven by shift registers, and dc/dc step-down power supply, and dashboard PCB.

University Outreach

MSU Science Festival

April 2024

Exhibitor

- Collaborated to prepare and staff a booth discussing our work on distributed communication and sensing technologies to a general audience, including a demonstrating of distributed time synchronization.

Introduce A Girl To Engineering Day

Feb. 2019, 2020

Volunteer

- Developed and ran interactive demonstrations for 4th-8th grade students to demonstrate electromagnetic scattering of different objects, and Doppler shift using a continuous-wave X-band Doppler radar and real-time visualization in Python

SpartaHack Hackathon

2015 – 2017

Founding Member, Organizer

- Helped with organization of the Spartahack hackathon for college and high school students from across the world

Spartan Hackers

Fall 2015

Founding Member

- Helped to organize and give talks at MSU on relevant computer science skills not taught in the classroom

Review Activities

Journal Reviewer

- IEEE Transactions on Radar Systems
- IEEE Transactions on Antennas and Propagation
- IEEE Antennas and Wireless Propagation Letters
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Signal Processing Letters
- IEEE Journal on Selected Areas in Communications

Conference Reviewer

- IEEE AP-S/URSI Symposium
- IEEE Vehicular Technology Conference
- European Conference on Antennas and Propagation (EUCaP)

Professional Society Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
 - IEEE Aerospace and Electronic Systems (AESS)
 - IEEE Antennas and Propagation Society (AP-S)
- IEEE Microwave Theory and Technology Society (MTT-S)
- IEEE Vehicular Technology Society (VTS)

Technical Skills

Programming Languages, Tools and Skills:

- Frequent Programming languages:
 - Python (Numpy, Numba, SciPy, PyQt, PyQtGraph, Matplotlib),
 - C++ (less frequently C)
- Software Defined Radios (SDRs)
 - GNU Radio 3.9+, UHD, Soapy SDR
 - Ettus X3x0, B2xx; some RFNoC experience
 - AD9361-based radios (B2xx, Litex M2SDR)
 - LMS6002D/LMS7002M-based (Wavelet-Labs uSDR)
- Parallel/high-performance computing,
- DevOps Skills:
 - Reproducible environment management (Docker, Pixi, Conda)
 - Git (GitLab CI tools), Computer Networking, Linux system administration
- Robotics Skills:
 - ROS, OpenCV

Design Tools and Skills:

- KiCad EDA, ANSYS HFSS, Keysight ADS, SPICE (circuit simulation), Autodesk AutoCAD, distributed element microwave circuits, antenna design, MVG StarLab antenna measurement